

Solutions to MTL106/MTL2006: Tutorial-2

Probability and Random Variables

Q1. A fair die is rolled twice. Let X denote the sum of the face values obtained on the two rolls, and let Y denote the absolute difference between the face values. Describe the sample space Ω . Determine whether X and Y are random variables on Ω . Justify your answer.

Solution:

- The sample space Ω consists of all possible outcomes of rolling a fair die twice.
- Each outcome can be represented as an ordered pair (i, j) where i and j are the results of the first and second rolls:

$$\Omega = \{(i, j) \mid i, j \in \{1, 2, 3, 4, 5, 6\}\}$$

- The size of the sample space is $|\Omega| = 36$.
- A random variable is formally defined as a measurable function from the sample space Ω to the real numbers $\mathbb{R}$.
- Since Ω is finite, the standard σ -algebra $\mathcal{F}$ associated with it is the power set (meaning every subset is measurable).
- The functions are defined as $X(i, j) = i + j$ and $Y(i, j) = |i - j|$.
- Because Ω is finite and equipped with the power set σ -algebra, *any* real-valued function defined on Ω is strictly measurable.
- Therefore, both X and Y satisfy the definition and are valid random variables.

Q2. Let $(\Omega, \mathcal{F}, P)$ be a probability space, and let $\mathcal{X}$ denote the set of all real-valued random variables defined on this probability space.

- Show that $\mathcal{X}$ is a vector space over $\mathbb{R}$.**
- Assume that the sample space Ω is finite. Determine a basis for the vector space $\mathcal{X}$ and hence find its dimension.**

Solution:

- To be a vector space, $\mathcal{X}$ must be closed under scalar multiplication and vector addition.
 - First, for any $X \in \mathcal{X}$ and $a \in \mathbb{R}$, the scalar multiple aX is clearly measurable.
 - Second, we must prove that if $X_1, X_2 \in \mathcal{X}$, their sum $X_1 + X_2$ is also a random variable.

- A function X is a random variable if for any $c \in \mathbb{R}$, the event set $\{\omega \in \Omega \mid X(\omega) \leq c\}$ belongs to the σ -algebra $\mathcal{F}$.
- To show $X_1 + X_2$ is a random variable, we must prove $\{\omega \mid X_1(\omega) + X_2(\omega) \leq c\} \in \mathcal{F}$.
- It is easier to first evaluate the complement event (strictly greater than): $\{\omega \mid X_1(\omega) + X_2(\omega) > c\}$.
- This strict inequality can be rearranged as: $X_1(\omega) > c - X_2(\omega)$.
- Because the rational numbers $\mathbb{Q}$ are countable and dense in $\mathbb{R}$, there must exist a rational number q strictly between these two values: $X_1(\omega) > q > c - X_2(\omega)$.
- This translates to two simultaneous conditions: $X_1(\omega) > q$ and $X_2(\omega) > c - q$.
- We can express the complement event as a countable union over all rational numbers $q \in \mathbb{Q}$:

$$\{\omega \mid X_1 + X_2 > c\} = \bigcup_{q \in \mathbb{Q}} (\{\omega \mid X_1 > q\} \cap \{\omega \mid X_2 > c - q\})$$

- Since X_1 and X_2 are random variables, the sets $\{\omega \mid X \leq x\} \in \mathcal{F}$, meaning their complements $\{\omega \mid X > x\}$ are also in $\mathcal{F}$.
- Therefore, the sets in the inner intersection belong to $\mathcal{F}$. The intersection of two sets in $\mathcal{F}$ is in $\mathcal{F}$.
- Because $\mathbb{Q}$ is countable, the countable union of these sets is also in $\mathcal{F}$. Thus, the complement event $\{\omega \mid X_1 + X_2 > c\} \in \mathcal{F}$.
- Finally, the complement of this entire set, which is exactly $\{\omega \mid X_1 + X_2 \leq c\}$, must also belong to $\mathcal{F}$.
- This proves $X_1 + X_2$ is a random variable. Since linear combinations are measurable, $\mathcal{X}$ is a vector space.

ii) Let the finite sample space be $\Omega = \{\omega_1, \omega_2, \dots, \omega_n\}$.

- For each outcome, define an indicator function $I_{\{\omega_k\}}(\omega)$ which equals 1 if $\omega = \omega_k$ and 0 otherwise.
- Any random variable $X \in \mathcal{X}$ can be expressed uniquely as a linear combination of these indicator functions:

$$X(\omega) = \sum_{k=1}^n X(\omega_k) I_{\{\omega_k\}}(\omega)$$

- The set of functions $\mathcal{B} = \{I_{\{\omega_1\}}, \dots, I_{\{\omega_n\}}\}$ spans the space and is linearly independent.
- Thus, $\mathcal{B}$ forms a basis for $\mathcal{X}$, and its dimension is exactly $n = |\Omega|$.

Q3. Determine whether each of the following functions is a cumulative distribution function (CDF). Justify your answer.

a) $F(x) = \begin{cases} 0 & x < 0 \\ x & 0 \leq x \leq 1/2 \\ 1 & x > 1/2 \end{cases}$

b) $F(x) = \frac{1}{\pi} \tan^{-1}(x), -\infty < x < \infty$

c) $F(x) = \begin{cases} 0 & x < 0 \\ 1 - e^{-x} & 0 \leq x < \infty \end{cases}$

Solution:

- A valid CDF must be non-decreasing, right-continuous, and satisfy $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.

a) **Not a CDF.**

- We check for right-continuity at the boundary $x = 1/2$.
- The function evaluates to $F(1/2) = 1/2$.
- However, the limit from the right is $\lim_{x \rightarrow 1/2^+} F(x) = 1$.
- Since $F(1/2) \neq \lim_{x \rightarrow 1/2^+} F(x)$, the function is not right-continuous.

b) **Not a CDF.**

- We check the infinite limits of the function.
- As $x \rightarrow \infty$, $\lim_{x \rightarrow \infty} \frac{1}{\pi} \tan^{-1}(x) = \frac{1}{\pi} \left(\frac{\pi}{2}\right) = \frac{1}{2}$, which is not 1.
- As $x \rightarrow -\infty$, the limit goes to $-1/2$, which is not 0.

c) **Is a CDF.**

- This is the standard cumulative distribution function of an Exponential random variable with $\lambda = 1$.
- It is perfectly right-continuous and strictly increasing for $x \geq 0$.
- The lower limit evaluates to $F(-\infty) = 0$.
- The upper limit evaluates to $\lim_{x \rightarrow \infty} (1 - e^{-x}) = 1 - 0 = 1$.

Q4. Let F and G be CDFs.

(a) **Show that $H(x) = \lambda F(x) + (1 - \lambda)G(x)$, $0 \leq \lambda \leq 1$ is also a CDF.**

Solution:

- $H(x)$ is a convex combination of two non-decreasing functions, so it is inherently non-decreasing.
- Being a linear combination of right-continuous functions, it is also right-continuous.
- For the upper limit: $\lim_{x \rightarrow \infty} H(x) = \lambda(1) + (1 - \lambda)(1) = \lambda + 1 - \lambda = 1$.
- For the lower limit: $\lim_{x \rightarrow -\infty} H(x) = \lambda(0) + (1 - \lambda)(0) = 0$.
- Thus, it meets all criteria and is a valid CDF.

(b) **Determine whether $H(x) = F(x)G(x)$ is necessarily a CDF.**

Solution: Yes.

- Since $F, G \geq 0$ and both are non-decreasing, their product is mathematically non-decreasing.
- The product of two right-continuous functions remains right-continuous.
- The infinite limits are valid: $0 \times 0 = 0$ as $x \rightarrow -\infty$, and $1 \times 1 = 1$ as $x \rightarrow \infty$.

(c) **Determine whether $H(x) = F(x) + (1 - F(x)) \log(1 - F(x))$ is necessarily a CDF.**

Solution: Yes.

- Let $y = F(x) \in [0, 1]$ and define an auxiliary function $h(y) = y + (1 - y) \log(1 - y)$.

- We check if $h(y)$ is non-decreasing by finding its derivative: $h'(y) = 1 - \log(1-y) - 1 = -\log(1-y)$.
- Since $0 \leq y \leq 1$, we know $1-y \leq 1$, meaning $\log(1-y) \leq 0$. Thus, $h'(y) \geq 0$.
- Since h is continuous and non-decreasing, and F is a right-continuous, non-decreasing CDF, their composition $H(x) = h(F(x))$ is right-continuous and non-decreasing.
- For the lower limit: As $x \rightarrow -\infty$, $F(x) \rightarrow 0$, which implies $h(0) = 0$.
- For the upper limit: As $x \rightarrow \infty$, $F(x) \rightarrow 1$, which implies $\lim_{y \rightarrow 1} h(y) = 1 + \lim_{y \rightarrow 1} (1-y) \log(1-y) = 1 + 0 = 1$.

Q5. Let f and g be probability density functions (PDFs).

i) **Show that $h(x) = \lambda f(x) + (1-\lambda)g(x)$, $0 \leq \lambda \leq 1$ is a density function.**

Solution:

- First, a PDF must be non-negative. $h(x) \geq 0$ since $f(x), g(x) \geq 0$ and the scalar $\lambda \in [0, 1]$.
- Second, a PDF must integrate to 1.
- $\int_{-\infty}^{\infty} h(x)dx = \lambda \int f(x)dx + (1-\lambda) \int g(x)dx$
- Since f and g are PDFs, their integrals are 1: $\lambda(1) + (1-\lambda)(1) = 1$.
- Thus, it satisfies both conditions of a valid PDF.

ii) **Determine whether $h(x) = f(x)g(x)$ is necessarily a PDF.**

Solution: No.

- The product of two PDFs does not necessarily integrate to 1.
- Counterexample: Let $f(x) = 2$ and $g(x) = 2$ for the interval $0 \leq x \leq 1/2$.
- Both are valid uniform PDFs that integrate to 1.
- However, their product is $h(x) = f(x)g(x) = 4$.
- Integrating this product gives $\int_0^{1/2} 4 dx = 2$, which is not equal to 1.

Q6. Let X be a continuous random variable with PDF $f(x) = \alpha + \beta x^2$ for $0 \leq x \leq 2$, and 0 otherwise. If $E[X] = 4/3$, determine α and β .

Solution:

- We formulate a system of equations using two absolute conditions: the PDF must integrate to 1, and the expected value is known.
- **Condition 1: Integration to 1**

$$\int_0^2 (\alpha + \beta x^2) dx = \left[\alpha x + \frac{\beta x^3}{3} \right]_0^2 = 2\alpha + \frac{8}{3}\beta = 1$$

- **Condition 2: Expected value is 4/3**

$$E[X] = \int_0^2 x(\alpha + \beta x^2) dx = \int_0^2 (\alpha x + \beta x^3) dx = \left[\frac{\alpha x^2}{2} + \frac{\beta x^4}{4} \right]_0^2 = 2\alpha + 4\beta = \frac{4}{3}$$

- We now have a linear system of two equations:

$$\begin{aligned} 2\alpha + \frac{8}{3}\beta &= 1 \\ 2\alpha + 4\beta &= \frac{4}{3} \end{aligned}$$

- Subtracting the first equation from the second eliminates α :

$$\left(4 - \frac{8}{3}\right)\beta = \frac{4}{3} - 1 \implies \left(\frac{4}{3}\right)\beta = \frac{1}{3} \implies \beta = \frac{1}{4}$$

- Substitute $\beta = 1/4$ back into the second equation:

$$2\alpha + 4\left(\frac{1}{4}\right) = \frac{4}{3} \implies 2\alpha + 1 = \frac{4}{3} \implies 2\alpha = \frac{1}{3} \implies \alpha = \frac{1}{6}$$

- Final values: $\alpha = 1/6$, $\beta = 1/4$.

Q7. Let X be a non-negative RV with CDF $F_X(x) = \frac{x}{2\pi}$ for $0 \leq x < 2\pi$, 0 for $x < 0$ and 1 for $x \geq 2\pi$. Find the CDF of $Y = X^2$. Determine whether X and Y are continuous RVs.

Solution:

- To find the CDF of Y , we evaluate $F_Y(y) = P(Y \leq y) = P(X^2 \leq y)$.
- We break this down into three distinct ranges for y :
- **Range 1** ($y < 0$): Since squares cannot be negative, $F_Y(y) = 0$.
- **Range 2** ($0 \leq y < 4\pi^2$): Since X is non-negative ($X \geq 0$), the probability becomes:

$$P(X^2 \leq y) = P(X \leq \sqrt{y}) = F_X(\sqrt{y}) = \frac{\sqrt{y}}{2\pi}$$

- **Range 3** ($y \geq 4\pi^2$): In this range, $\sqrt{y} \geq 2\pi$. The original CDF F_X is maxed out at 1, so $F_Y(y) = 1$.
- The final piecewise CDF of Y is:

$$F_Y(y) = \begin{cases} 0 & y < 0 \\ \frac{\sqrt{y}}{2\pi} & 0 \leq y < 4\pi^2 \\ 1 & y \geq 4\pi^2 \end{cases}$$

- X is a continuous RV because its CDF $F_X(x)$ has no jump discontinuities and is everywhere continuous.
- Y is also a continuous RV because its CDF $F_Y(y)$ evaluates to exactly $\frac{\sqrt{4\pi^2}}{2\pi} = 1$ at the upper boundary and 0 at the lower boundary, meaning it is continuous everywhere.

Q8. Let X be a random variable with CDF: $F(x) = 0$ for $x < 0$, $x/2$ for $0 \leq x < 1$, $2/3$ for $1 \leq x < 2$, $11/12$ for $2 \leq x < 3$, and 1 for $x \geq 3$. Compute probabilities:

Solution:

- i) **Computing $P(X > 1/2)$:**

$$P(X > 1/2) = 1 - P(X \leq 1/2) = 1 - F(1/2) = 1 - \frac{1/2}{2} = 1 - \frac{1}{4} = \frac{3}{4}$$

- ii) **Computing $P(2 < X \leq 4)$:**

$$P(2 < X \leq 4) = F(4) - F(2) = 1 - \frac{11}{12} = \frac{1}{12}$$

- iii) **Computing $P(X < 3)$:**

$$P(X < 3) = \lim_{x \rightarrow 3^-} F(x) = \frac{11}{12}$$

- iv) **Computing $P(X = 1)$:**

$$P(X = 1) = F(1) - \lim_{x \rightarrow 1^-} F(x) = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$$

Q9. Let X satisfy $P(0 \leq X \leq c) = 1$. Show that $\text{Var}(X) \leq c^2/4$.

Solution:

- Since X falls in the interval $[0, c]$ with probability 1, we know that $X \geq 0$ and $(X - c) \leq 0$.
- Because one term is positive and the other is negative (or zero), their product is strictly non-positive: $X(X - c) \leq 0$.
- Expanding this expression gives: $X^2 - cX \leq 0$.
- Taking the expected value of both sides preserves the inequality: $\mathbb{E}[X^2] - c\mathbb{E}[X] \leq 0$, which means $\mathbb{E}[X^2] \leq c\mathbb{E}[X]$.
- Let the mean of X be $\mu = \mathbb{E}[X]$.
- The standard formula for variance is $\text{Var}(X) = \mathbb{E}[X^2] - \mu^2$.
- Substituting our inequality into the variance formula yields: $\text{Var}(X) \leq c\mu - \mu^2$.
- The expression $c\mu - \mu^2$ forms a downward-opening quadratic parabola in terms of μ .
- To find the maximum possible value, we locate its vertex by taking the derivative and setting it to 0, which occurs at $\mu = c/2$.
- The maximum value of the parabola is therefore $c(c/2) - (c/2)^2 = c^2/2 - c^2/4 = c^2/4$.
- Hence, the variance is bounded by: $\text{Var}(X) \leq c^2/4$.

Q10. Let X be positive with PDF $f_X(x)$. Define $U = X/(1 + X)$. Determine the PDF of U .

Solution:

- **Define the bounds:** Let $u = g(x) = \frac{x}{1+x}$. Since X is strictly positive ($x > 0$), the valid range for U is $0 < u < 1$.

- **Find the inverse function:** We solve for x in terms of u .

$$u(1+x) = x \implies u+ux = x \implies u = x-ux \implies u = x(1-u) \implies x = g^{-1}(u) = \frac{u}{1-u}$$

- **Calculate the derivative:** Differentiate the inverse function with respect to u .

$$\frac{dx}{du} = \frac{(1)(1-u) - (u)(-1)}{(1-u)^2} = \frac{1-u+u}{(1-u)^2} = \frac{1}{(1-u)^2}$$

- **Apply the formula for the PDF of a transformed random variable:** The PDF of a transformed random variable is $f_U(u) = f_X(g^{-1}(u)) \left| \frac{dx}{du} \right|$.
- **Final PDF:** Substituting our derived components, we obtain:

$$f_U(u) = \begin{cases} \frac{1}{(1-u)^2} f_X\left(\frac{u}{1-u}\right) & \text{for } 0 < u < 1 \\ 0 & \text{otherwise} \end{cases}$$

Q11. Let X have PDF $f(x) = \beta 5^\beta / x^{\beta+1}$ for $x \geq 5$, and 0 otherwise ($\beta > 0$). Show $E[X^n]$ exists iff $n < \beta$. Assume $\beta > 2$, find mean and variance.
Solution:

- The n -th moment expectation is defined by the integral:

$$E[X^n] = \int_5^\infty x^n \frac{\beta 5^\beta}{x^{\beta+1}} dx = \beta 5^\beta \int_5^\infty x^{n-\beta-1} dx$$

- This improper integral converges to a finite value if and only if the exponent is strictly less than -1 .
- Solving the inequality $n - \beta - 1 < -1 \implies n - \beta < 0 \implies n < \beta$. This proves the condition.
- **Find the Mean ($n = 1$):** Assuming $\beta > 2$ (so it converges):

$$E[X] = \beta 5^\beta \int_5^\infty x^{-\beta} dx = \beta 5^\beta \left[\frac{x^{1-\beta}}{1-\beta} \right]_5^\infty = 0 - \frac{\beta 5^\beta 5^{1-\beta}}{1-\beta} = \frac{5\beta}{\beta-1}$$

- **Find the Second Moment ($n = 2$):**

$$E[X^2] = \beta 5^\beta \int_5^\infty x^{1-\beta} dx = \beta 5^\beta \left[\frac{x^{2-\beta}}{2-\beta} \right]_5^\infty = 0 - \frac{\beta 5^\beta 5^{2-\beta}}{2-\beta} = \frac{25\beta}{\beta-2}$$

- **Calculate the Variance:** Using $\text{Var}(X) = E[X^2] - (E[X])^2$:

$$\text{Var}(X) = \frac{25\beta}{\beta-2} - \frac{25\beta^2}{(\beta-1)^2}$$

Q12. Define $Y = \alpha X + \beta$. Determine PGF of Y in terms of $G_X(\cdot)$.
Solution:

- The probability generating function (PGF) of Y is defined by the expectation $G_Y(s) = E[s^Y]$.
- Substitute the definition of Y into the exponent:

$$G_Y(s) = E[s^{\alpha X + \beta}]$$

- Use exponent algebra to separate the terms:

$$G_Y(s) = E[s^\beta \cdot s^{\alpha X}] = E[s^\beta (s^\alpha)^X]$$

- Since s^β is a constant with respect to the expectation, pull it outside:

$$G_Y(s) = s^\beta E[(s^\alpha)^X]$$

- By definition, the PGF of X evaluated at some input z is $E[z^X] = G_X(z)$.
- Treating $z = s^\alpha$, we substitute this back in to get the final relationship:

$$G_Y(s) = s^\beta G_X(s^\alpha)$$

Q13. Let X be a discrete RV with PGF $G_X(s) = \frac{s}{3}(2 + s^2)$. Determine PMF.

Solution:

- The given probability generating function is $G_X(s) = \frac{s}{3}(2 + s^2)$.
- Expanding this expression algebraically yields: $G_X(s) = \frac{2}{3}s + \frac{1}{3}s^3$.
- By definition, a probability generating function takes the form $G_X(s) = \sum_{k=0}^{\infty} P(X = k)s^k$.
- Therefore, the coefficient of each s^k term directly corresponds to the probability $P(X = k)$.
- Matching our expanded coefficients to the corresponding powers of s , the PMF is:

- $P(X = 1) = 2/3$
- $P(X = 3) = 1/3$
- $P(X = x) = 0$ for all other x .

Q14. $f_X(x) = e^{-2x} + \frac{1}{2}e^{-x}$ for $x > 0$. Determine MGF, $E[X]$, and $E[X^2]$.

Solution:

i) **Determine the MGF:**

- The Moment Generating Function is defined as $M_X(t) = E[e^{tX}]$.
- Setting up the integral: $M_X(t) = \int_0^{\infty} e^{tx}(e^{-2x} + 0.5e^{-x})dx$.
- Combining the exponents: $\int_0^{\infty} (e^{(t-2)x} + 0.5e^{(t-1)x})dx$.
- Evaluating the integral (which converges for $t < 1$):

$$M_X(t) = \left[\frac{e^{(t-2)x}}{t-2} + \frac{0.5e^{(t-1)x}}{t-1} \right]_0^{\infty} = 0 - \left(\frac{1}{t-2} + \frac{0.5}{t-1} \right) = \frac{1}{2-t} + \frac{1}{2(1-t)}$$

ii) **Compute Moments via Derivatives:**

- First derivative: $M'(t) = (2-t)^{-2} + \frac{1}{2}(1-t)^{-2}$.
- Second derivative: $M''(t) = 2(2-t)^{-3} + (1-t)^{-3}$.
- The first moment is $E[X] = M'(0) = \frac{1}{2^2} + \frac{1}{2(1^2)} = \frac{1}{4} + \frac{1}{2} = \frac{3}{4}$.
- The second moment is $E[X^2] = M''(0) = \frac{2}{2^3} + \frac{1}{1^3} = \frac{2}{8} + 1 = \frac{1}{4} + 1 = \frac{5}{4}$.

Q15. Determine whether the MGF of a Cauchy RV exists.

Solution:

- The probability density function of a standard Cauchy random variable is $f(x) = \frac{1}{\pi(1+x^2)}$.
- The formulation for its Moment Generating Function is $M_X(t) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\pi(1+x^2)} dx$.
- Consider the behavior of the integrand for any positive $t > 0$. As $x \rightarrow \infty$, the numerator e^{tx} grows exponentially.
- However, the denominator $\pi(1+x^2)$ grows only polynomially (quadratically).
- Because exponential growth vastly overtakes polynomial growth, the entire fraction approaches infinity rather than approaching zero.
- The exact same divergent behavior occurs for any negative $t < 0$ as $x \rightarrow -\infty$.
- Because the integrand does not decay to zero, the area under the curve is infinite.
- Thus, the integral diverges for all $t \neq 0$, meaning the MGF does not mathematically exist.

Q16. Let $m_X(t) = a + be^{2t} + ce^{4t}$. $E[X] = 3$, $\text{Var}(X) = 2$. Determine $E[2^X]$.

Solution:

- By definition of an MGF, it must evaluate to 1 when $t = 0$. Therefore, $m_X(0) = a + b + c = 1$.
- We find the moments by taking derivatives of the MGF and evaluating at $t = 0$.
- First derivative (Expectation): $E[X] = m'_X(0) = 2b + 4c = 3$.
- Second derivative: $E[X^2] = m''_X(0) = 4b + 16c$.
- We can find the numeric value for $E[X^2]$ using the variance formula: $E[X^2] = \text{Var}(X) + (E[X])^2 = 2 + 3^2 = 11$.
- Equating these gives our second constraint: $4b + 16c = 11$.
- We now solve the system of equations. Multiply $2b + 4c = 3$ by 2 to get $4b + 8c = 6$.
- Subtract this from $4b + 16c = 11$ to eliminate b : $8c = 5 \implies c = 5/8$.
- Substitute c back to find b : $2b + 4(5/8) = 3 \implies b = 1/4$.
- Substitute b and c to find a : $a = 1 - 1/4 - 5/8 = 1/8$.
- Now, we evaluate the target expectation. Notice that $E[2^X] = E[e^{X \ln 2}]$, which perfectly matches the definition of the MGF evaluated at $t = \ln 2$.

- Substitute $t = \ln 2$ into the original MGF equation:

$$m_X(\ln 2) = a + be^{2\ln 2} + ce^{4\ln 2} = a + b(e^{\ln 4}) + c(e^{\ln 16}) = a + 4b + 16c$$

- Plugging in our found constants: $E[2^X] = \frac{1}{8} + 4\left(\frac{1}{4}\right) + 16\left(\frac{5}{8}\right) = \frac{1}{8} + 1 + 10 = \frac{89}{8} = 11.125$.

Q17. Show $1 - |\varphi_X(2u)|^2 \leq 4(1 - |\varphi_X(u)|^2)$.

Solution:

- We use the method of symmetrization. Let Y be an independent, identically distributed (i.i.d.) copy of X .
- The squared magnitude of the complex characteristic function can be written as:

$$|\varphi_X(u)|^2 = \varphi_X(u)\varphi_X(-u) = \mathbb{E}[e^{iuX}]\mathbb{E}[e^{-iuY}]$$

- Because X and Y are independent, the product of their individual expectations perfectly equals the expectation of their product:

$$|\varphi_X(u)|^2 = \mathbb{E}[e^{iuX} \cdot e^{-iuY}] = \mathbb{E}[e^{iu(X-Y)}]$$

- Define a new variable $Z = X - Y$. Since X and Y are i.i.d., Z is perfectly symmetric around 0.
- Expanding the expectation using the complex exponential definition gives: $\mathbb{E}[e^{iuZ}] = \mathbb{E}[\cos(uZ)] + i\mathbb{E}[\sin(uZ)]$.
- Because Z is symmetric and the sine function is odd, the expected value of the imaginary part cancels out entirely: $\mathbb{E}[\sin(uZ)] = 0$.
- This strictly isolates the real part: $|\varphi_X(u)|^2 = \mathbb{E}[\cos(uZ)]$.
- Substitute this newly found relationship into both sides of our target inequality to yield an equivalent statement:

$$1 - \mathbb{E}[\cos(2uZ)] \leq 4(1 - \mathbb{E}[\cos(uZ)])$$

- To prove this statement holds, we evaluate the underlying trigonometric functions for any general angle θ . Using the trigonometric identity $\cos(2\theta) = 2\cos^2(\theta) - 1$:

$$1 - \cos(2\theta) = 1 - (2\cos^2\theta - 1) = 2 - 2\cos^2\theta = 2(1 - \cos^2\theta)$$

- Factoring this difference of squares yields: $2(1 - \cos\theta)(1 + \cos\theta)$.
- Because the absolute maximum possible value of $\cos\theta$ is 1, the maximum value of the term $(1 + \cos\theta)$ is exactly 2.
- Because the term $(1 - \cos\theta)$ is always non-negative, substituting the maximum value establishes a strict inequality:

$$2(1 - \cos\theta)(1 + \cos\theta) \leq 2(1 - \cos\theta)(2) = 4(1 - \cos\theta)$$

- Since $1 - \cos(2\theta) \leq 4(1 - \cos\theta)$ is unconditionally mathematically true for all real θ , we can set $\theta = uZ$.

- Because the inequality is valid pointwise, taking the overall expectation (the weighted average) of both sides preserves the inequality:

$$\mathbb{E}[1 - \cos(2uZ)] \leq \mathbb{E}[4(1 - \cos(uZ))]$$

- This translates perfectly back to our target inequality: $1 - |\varphi_X(2u)|^2 \leq 4(1 - |\varphi_X(u)|^2)$.

Q18. Define $Y = F_X(X)$ where X is continuous. Determine distribution and characteristic function of Y .

Solution:

- The CDF of Y is defined by evaluating the probability $F_Y(y) = P(Y \leq y) = P(F_X(X) \leq y)$.
- Since $F_X(x)$ represents probabilities, the random variable Y can strictly only take values in $[0, 1]$.
- For $y \in (0, 1)$, we define the generalized inverse (also known as the quantile function) as $F_X^{-1}(y) = \inf\{x \in \mathbb{R} : F_X(x) \geq y\}$.
- **Case 1: F_X is strictly increasing (no flat spots).**
 - In this case, the standard mathematical inverse exists, and F_X is a strict one-to-one mapping.
 - We can directly apply the inverse function to both sides of the inequality inside the probability calculation:

$$P(F_X(X) \leq y) = P(X \leq F_X^{-1}(y)) = F_X(F_X^{-1}(y)) = y$$

- **Case 2: F_X is not strictly increasing (has flat spots).**
 - If F_X is constant over a specific interval, the generalized inverse returns the absolute smallest x that reaches the probability limit y .
 - An interval where F_X is completely flat heavily implies that the probability density in that interval is exactly 0.
 - It is therefore impossible for the random variable X to ever take values strictly inside the flat region.
 - Because X cannot land in the non-invertible gap anyway, the event $\{F_X(X) \leq y\}$ remains perfectly mathematically equivalent to $\{X \leq F_X^{-1}(y)\}$.
 - Even without a strict 1-to-1 inverse mapping, the cumulative probability evaluates identically:

$$P(F_X(X) \leq y) = P(X \leq F_X^{-1}(y)) = F_X(F_X^{-1}(y)) = y$$

- In all cases, $F_Y(y) = y$ for all $0 \leq y \leq 1$.
- Because $F_Y(y) = y$ is the exact CDF of a standard uniform distribution, it means $Y \sim \text{Uniform}(0, 1)$.
- The characteristic function of a $\text{Uniform}(0, 1)$ RV is evaluated via integration as:

$$\varphi_Y(t) = \int_0^1 e^{ity} \cdot 1 \, dy = \left[\frac{e^{ity}}{it} \right]_0^1 = \frac{e^{it} - 1}{it} \text{ for } t \neq 0, \text{ and } \varphi_Y(0) = 1.$$